

Motor Gasoline BS-IV (IS 2796:2008 Amd.1) ? Grade MG 91: Specification Summary

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Specification matrix (MG 91) - values transcribed 1:1 from the source table

- i. Colour - As decided by refiners & marketers - [Visual] (typical Orange)
- ii. Density at 15°C - 720-775 kg/m³ [IS 1448, P:16] (typical 751)
- iii. E70 (recovery up to 70°C) - 10-45 % vol [IS 1448, P:18] (typical 25)
- iv. E100 (recovery up to 100°C) - 40-70 % vol (typical 55)
- v. E150 (recovery up to 150°C) - 75 min % vol (typical 89)
- vi. Final Boiling Point - 210 max °C (typical 180)
- vii. Residue - 2 max % vol (typical 0.5)
- viii. Research Octane Number (RON) - 91 min - [IS 1448, P:27] (typical 91.3)
- ix. Motor Octane Number (MON) - 81 min - [IS 1448, P:26] (typical 83.5)
- x. Existent Gum - 40 max g/m³ [IS 1448, P:147] (typical 18)
- xi. Sulphur, total - 50 max mg/kg [IS 1448, P:34] (typical 25)
- xii. Lead Content (as Pb) - 0.005 max g/l [IS 1448, P:80] (typical < 0.001)
- xiii. Reid Vapour Pressure at 38°C - 60 kPa [IS 1448, P:39] (typical 47)
- xiv. Vapour Lock Index, Summer / Winter - 750 / 950 max - (typical 620 / 650)
- xv. Benzene - 1.0 max % vol [ASTM D 3606] (typical 0.50)
- xvi. Copper strip corrosion (3h at 50°C) - Not more than 1 class [IS 1448, P:15] (typical Class 1)
- xvii. Olefin Content - 21 max % vol [P:23] (typical 1.3)
- xviii. Oxidation Stability - 360 min minutes [P:28] (typical Above 360)
- xix. Aromatic Content - 35 max % vol [P:23] (typical 33.5)
- xx. Oxygen Content - 2.7 max % mass [ISO 1601] (typical NA*)
- xxi. Ethanol (oxygenate) - 5 max % vol [ASTM D 4815] (typical -)

Engineering summary

RON 91 min / MON 81 min; sulphur 50 mg/kg max; lead 0.005 g/l max; benzene 1.0 % vol max.

Volatility bounded by E70 10-45, E100 40-70, RVP 60 kPa, VLI 750/950 max (summer/winter).

Typical analysis sits inside every limit (RON 91.3, sulphur 25, lead < 0.001).